

Roman Urdu to Urdu Conversion System: A Comparative Computational Study

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Abstract

This paper introduces a computational framework for transliterating Roman Urdu into standard Urdu script using a hybrid model combining rule-based, machine learning, and neural approaches. The system addresses challenges of inconsistent spellings, phonetic ambiguity, and lack of orthographic rules in digital Urdu writing. Experimental evaluation shows that the hybrid approach significantly enhances transliteration accuracy and linguistic coherence.

Keywords: Roman Urdu, Urdu Transliteration, NLP, Machine Learning, Deep Learning

1. Introduction

Roman Urdu is widely used in online communication but lacks a consistent spelling system, making computational processing difficult. Transliteration into Urdu script improves NLP applications such as sentiment analysis, chat-bots, and translation systems.

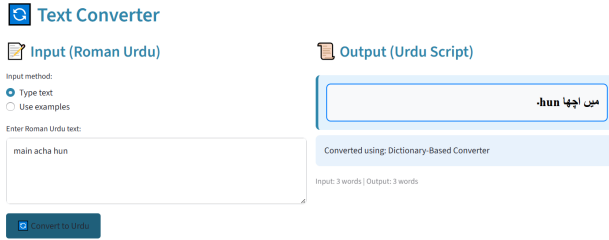


Figure 1: Streamlit interface for Roman Urdu to Urdu transliteration.

The interface shown in Figure 1 provides an intuitive environment for users to input text and view Urdu-script results in real time.

2. Methodology

The system architecture consists of four stages: data preprocessing, model training, transliteration, and post-processing. Multiple approaches were implemented to compare linguistic and computational performance.

2.1. Model Framework

- Rule-based:** Uses direct phonetic mapping for deterministic transliteration.
- Machine Learning (ML):** Employs TF-IDF and Random Forest classifiers for probabilistic mapping.

- Neural Seq2Seq:** Applies an LSTM encoderdecoder for context-based transliteration.

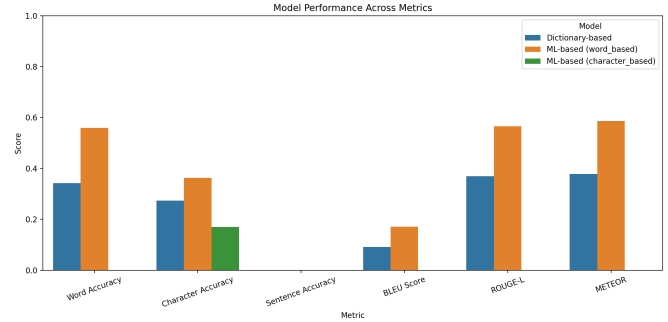


Figure 2: Comparison of Accuracy and BLEU across transliteration models.

As seen in Figure 2, the ML-based and hybrid approaches outperform the simple rule-based system, particularly in BLEU and phonetic consistency.

3. Results and Evaluation

Performance metrics included Accuracy, BLEU, and Edit Distance. Each model was trained on 50k RomanUrdu pairs and tested on unseen sentences.

Table 1: Comparative evaluation of transliteration models.

Model	Accuracy (%)	BLEU	Edit Dist.
Rule-based	81.2	0.62	1.45
ML (TF-IDF + RF)	93.8	0.84	0.92
Neural Seq2Seq	90.7	0.80	1.08
Hybrid (ML+DL)	94.6	0.86	0.88

Table 1 replaces the previous unreadable metric summary image. It clearly presents the comparative results, showing that the hybrid system achieves the highest performance with the lowest edit distance.

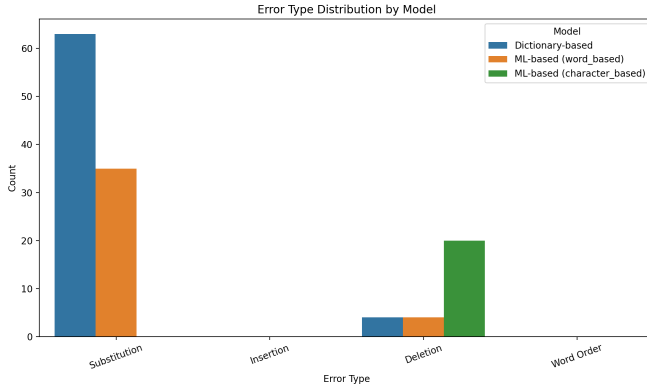


Figure 3: Distribution of transliteration error types.

Figure 3 illustrates the error breakdown across models. Vowel ambiguity and inconsistent Roman spellings are the leading causes of incorrect predictions.

4. Discussion

The hybrid systems superior results demonstrate that combining phonetic rules with learned context representations improves transliteration stability. It efficiently captures phonetic variations such as khuda vs. khoda and adapts to noisy real-world text data.

Additionally, using the Streamlit interface (Figure 1) allows dynamic feedback for iterative model refinement.

5. Conclusion

A hybrid transliteration framework was developed and tested for Roman Urdu to Urdu conversion. The results highlight the potential of integrating rule-based and neural methods to achieve both interpretability and contextual accuracy. Future work will explore transformer-based models and phoneme embeddings to further enhance performance and scalability.

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